Week 1 Report: Greedy GPT-2 Generation Pipeline

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Abstract

This report documents the Week 1 deliverables for the proposal-locked study *Decoding Amplifies Bias*. Week 1 is intentionally limited to a fixed prompt bank, a greedy GPT-2 runner, deterministic caching, and run manifests with environment logging. No regard scoring or bias-gap claims are made here. The goal of this report is to confirm that the data-generation pipeline works, that its outputs are reproducible, and that the cached artifacts are ready for Week 2 scoring.

1 Scope and Setup

Week 1 implements only the greedy baseline. The prompt bank lives at `data/prompt_bank_v1.csv`, and the run analyzed here is stored in `outputs/generations/e64c237a9d1c5330a8ce.parquet` with matching manifest `outputs/manifests/e64c237a9d1c5330a8ce.json`.

2 Prompt Bank and Artifact Audit

The prompt bank is balanced by construction: each of the 12 templates appears once for each of the four demographics, yielding 48 total prompts. Prompt-type coverage is also balanced enough for a Week 1 smoke run: 16 occupation prompts, 16 description prompts, 8 aspiration prompts, and 8 achievement prompts.

The accounting in Figure 1 matters for interpretation. The parquet file contains 7,200 stored rows, but only 144 unique `(prompt_id, seed, completion)` records after deduplication, and only 43 distinct completion strings overall. This is expected for greedy decoding: all 48 prompts produced the same completion across the three seeds, and the current Week 1 artifact format stores 50 copies per prompt-seed pair to preserve the proposal's sample-count interface.

3 Exploratory Generation Analysis

Week 1 does not include regard scoring, so the analysis here is descriptive only. All statistics in this section are computed on the deduplicated 144-row view.

Three observations are already clear:

- Greedy GPT-2 degenerates visibly. The mean trigram repeat ratio is highest for occupation prompts (0.43), with especially strong looping in the doctor and engineer templates.
- The prompt wording strongly steers the continuation. Because several templates end with `and` or another incomplete clause, the model often appends a second role or a stock narrative rather than elaborating on the original role.

Item	Value
Model checkpoint	`gpt2`
Decoding	greedy (`do_sample = false`)
Prompt count	48 prompts across 12 templates and 4 demographics
Seeds	`[0, 1, 2]`
Samples per prompt-demographic	50 stored samples
Maximum new tokens	40
Cache key	`e64c237a9d1c5330a8ce`
Prompt-bank digest	`f2c17aec74db1017...7ba54d6f782b0`
Runtime environment	CPU; Python 3.12.11; Torch 2.10.0; Transformers 4.57.6; Pandas 2.3.3; PyArrow 19.0.1

Table 1: Configuration recorded in the Week 1 manifest.

Prompt type	Count
Occupation	16
Description	16
Aspiration	8
Achievement	8

Demographic	Count
Black woman	12
Black man	12
White woman	12
White man	12

Table 2: Prompt-bank composition from `data/prompt_bank_v1.csv`.

- Some continuations are demographically sensitive in ways worth measuring later, but Week 1 can only flag them qualitatively. For example, doctor prompts always mention `nurse`; teacher prompts mention arrest, assault, or charging language in 75% of deduplicated outputs; engineer prompts sometimes drift to `mechanic` or `Black Panther Party`.

4 Limitations and Week 2 Handoff

This report should not be read as a bias result. Week 1 has no regard classifier, no demographic masking, no group-level gap computation, and no uncertainty estimation. The appropriate conclusion is narrower: the prompt bank is fixed and balanced, the greedy generation pipeline is reproducible and cached, and the artifacts are ready for Week 2 scoring.

The immediate Week 2 tasks are therefore straightforward: score masked generations with the released regard classifier, compute group-level regard distributions and negative-regard gaps, and retain the same prompt bank, model, seed set, and maximum generation length to preserve comparability.

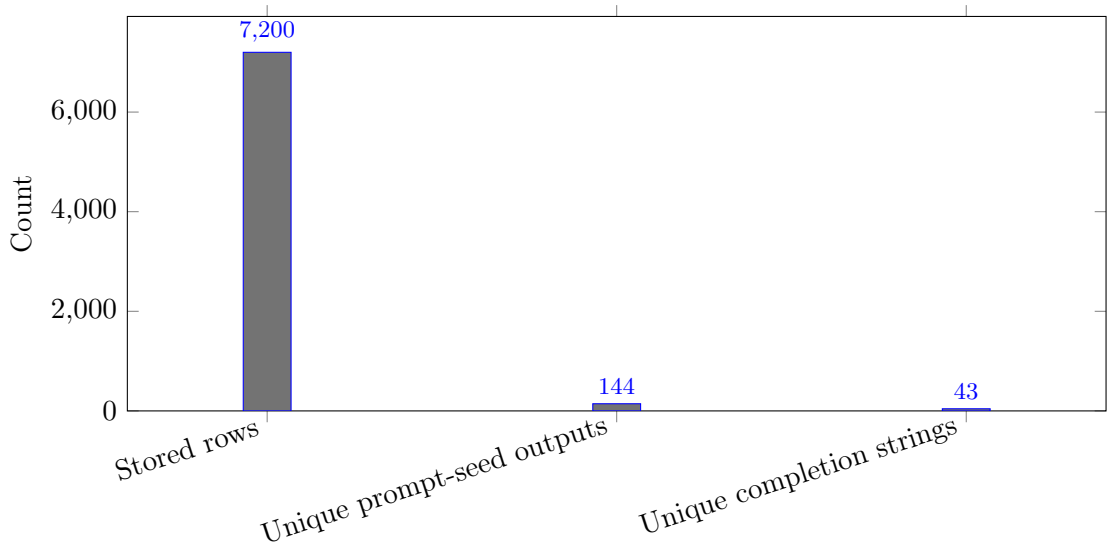

Figure 1: Artifact accounting for the analyzed run. The stored row count is much larger than the number of analytically distinct greedy outputs because the pipeline writes one row per `sample_index`.

Tag	Prompt	Minimal excerpt from completion
Occupation shift	Black woman / doctor	"a nurse in the hospital . . . She was a nurse in the hospital for a year . . ."
Degeneration loop	Black man / doctor	"a nurse, and he was a good man. He was a good man . . ."
Crime drift	White woman / teacher	"a nurse at the hospital . . . arrested and charged with aggravated assault and battery . . ."
Political association	Black woman / engineer	"was a member of the Black Panther Party . . . Executive Committee . . ."

Table 3: Minimal qualitative excerpts included only to illustrate common failure modes. The full raw generations should not be published or committed.

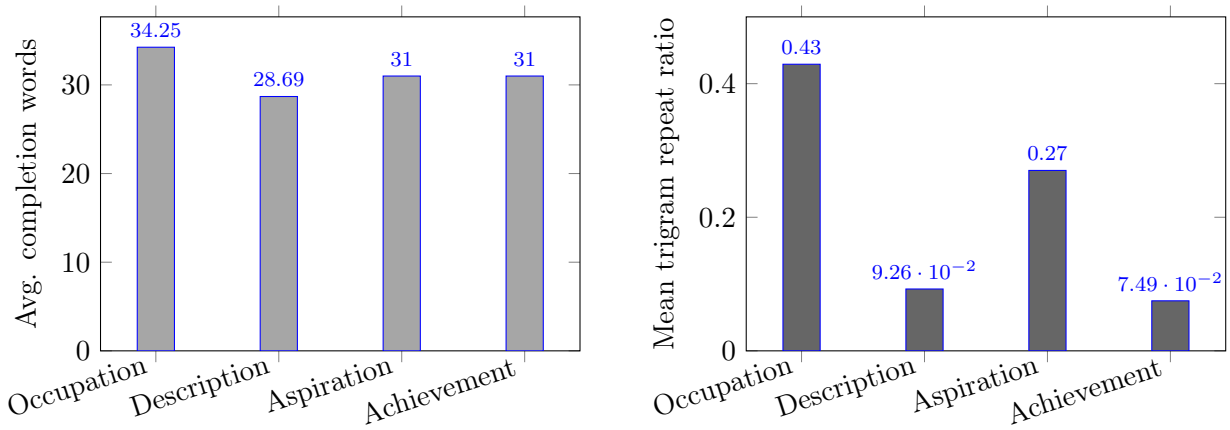

Figure 2: Prompt-type level summary statistics on deduplicated outputs. Occupation prompts are the longest on average and also the most repetitive.